

SEEM5020 Algorithms for Big Data

Dimension Reduction with the Johnson-Lindenstrauss Lemma

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Prevalence of Vector Data

Recent advances in deep learning encode various unstructured data, like text, video, and image, into high-dimensional vectors. Many real-world applications now directly deal with these vector data.

- YouTube video to vector data for video recommendation
- Airbnb maps their property descriptions to vectors for search ranking
- Alibaba represents products as vectors for product recommendation
- ...

Dimension Reduction

Definition 1 (Dimension reduction)

Given a set of points $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ in $\mathbb{R}^d$ (where d is the original high dimension), the goal of dimension reduction is to find a function $f : \mathbb{R}^d \rightarrow \mathbb{R}^k$ such that $k < d$, and each point $\mathbf{x}_i$ is mapped to a point in $\mathbb{R}^k$ with preservation of certain properties (e.g., pairwise distances) in the lower-dimensional space.

Why do we need dimension reduction?

- Efficiency: Reducing the dimensionality can speed up algorithms without significantly compromising the quality of results.
- Curse of dimensionality: High-dimensional spaces can be counterintuitive; algorithms can behave poorly as dimension increases.
- Storage: Reducing the storage space.
- ...

Preserving the Pairwise Euclidean Distance

Problem 1

Given a set of points $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ in $\mathbb{R}^d$, we want to find a function $f: \mathbb{R}^d \rightarrow \mathbb{R}^k$ such that $k < d$, and each point $\mathbf{x}_i$ is mapped to a point in $\mathbb{R}^k$, such that for any i, j we have:

$$(1 - \varepsilon) \|\mathbf{x}_i - \mathbf{x}_j\|_2 \leq \|f(\mathbf{x}_i) - f(\mathbf{x}_j)\|_2 \leq (1 + \varepsilon) \|\mathbf{x}_i - \mathbf{x}_j\|_2 \quad (1)$$

Applications:

- Approximate all-pair distances.
- Distance-based clustering like k -means, DBSCAN, Hierarchical Clustering.
- Linear-regression
- ...

The Johnson-Lindenstrauss Lemma [1]

Lemma 1 (JL Lemma)

There is a *linear mapping* function f that maps $\mathbf{x}_i$ ($1 \leq i \leq n$) with d dimensions to $\mathbf{x}'_i$ with $k = O(\frac{\log n}{\epsilon^2})$ dimensions that satisfies Equation (1).

Some tricks that we might start with based on our previous lectures:

- Random Gaussian projection
- Random Sign projection
- Random coordinate selection
- ...

The (ε, δ) -JL Property

Theorem 1 ((ε, δ)-JL Property)

Let $\mathbf{\Pi}$ be a $k \times d$ random matrix with each entry being a normalized random Gaussian variable, i.e., $\Pi_{i,j} \sim \frac{1}{\sqrt{k}} \mathcal{N}(0,1)$. Then by setting $f(\mathbf{x})$ as $f(\mathbf{x}) = \mathbf{\Pi} \cdot \mathbf{x}$, the following holds for an arbitrary vector $\mathbf{x}$:

$$(1 - \varepsilon) \|\mathbf{x}\|_2^2 \leq \|\mathbf{\Pi} \mathbf{x}\|_2^2 \leq (1 + \varepsilon) \|\mathbf{x}\|_2^2.$$

with $1 - \delta$ probability when $k = O(\frac{\log(1/\delta)}{\varepsilon^2})$.

Remark 1: The above theorem actually provides a tighter bound as required by Equation (1). It is simple to obtain the bound following that of Equation (1) by taking a square root on each side.

Remark 2: Given the above theorem, by setting $\delta' = \frac{\delta}{\binom{n}{2}}$ in Theorem 1, we can guarantee that Equation (1) holds for an arbitrary pair.

Analysis of the (ε, δ) -JL Property

Define $\mathbf{G} = \sqrt{k} \cdot \mathbf{\Pi}$, i.e., each entry in $\mathbf{G}$ is a Gaussian random variable following $\mathcal{N}(0, 1)$. Let

$$\mathbf{G} = \begin{pmatrix} \mathbf{g}_1^T \\ \mathbf{g}_2^T \\ \vdots \\ \mathbf{g}_k^T \end{pmatrix}$$

Then, for an arbitrary vector $\mathbf{x}$, let $\mathbf{w} = \mathbf{G} \cdot \mathbf{x}$ and w_i be the i -th entry in $\mathbf{w}$. Then, we have

$$w_i = \mathbf{g}_i^T \cdot \mathbf{x}.$$

Accordingly,

$$\begin{aligned} \|\mathbf{\Pi} \cdot \mathbf{x}\|_2^2 &= \left\| \frac{1}{\sqrt{k}} \mathbf{G} \cdot \mathbf{x} \right\|_2^2 = \frac{1}{k} \|(\mathbf{g}_1^T \mathbf{x}, \mathbf{g}_2^T \mathbf{x}, \dots, \mathbf{g}_k^T \mathbf{x})^T\|_2^2 \\ &= \frac{1}{k} \sum_{i=1}^k w_i^2. \end{aligned}$$

Analysis of the (ε, δ) -JL Property (Cont.)

Let $\mathbf{x} = (x_1, x_2, \dots, x_d)^T$ and $\mathbf{g}_i = (g_{i,1}, g_{i,2}, \dots, g_{i,d})^T$. For w_i , then it is

$$w_i = x_1 \cdot g_{i,1} + x_2 \cdot g_{i,2} + \dots + x_d \cdot g_{i,d}.$$

Notice that $g_{i,j}$ is a Gaussian random variable from $\mathcal{N}(0, 1)$.

Lemma 2 (Gaussian Distribution Properties (i))

$X \sim \mathcal{N}(0, \sigma^2)$. Then, given a positive a , $a \cdot X \sim \mathcal{N}(0, a^2 \cdot \sigma^2)$.

Lemma 3 (Gaussian Distribution Properties (ii))

$X \sim \mathcal{N}(\mu_1, \sigma_1^2), Y \sim \mathcal{N}(\mu_2, \sigma_2^2) \rightarrow Z = X + Y \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$.

Therefore, $x_j \cdot g_{i,j}$ is a Gaussian random variable following the distribution $\mathcal{N}(0, x_j^2)$ and w_j is a Gaussian random variable following the distribution $\mathcal{N}(0, x_1^2 + x_2^2 + \dots + x_d^2) = \mathcal{N}(0, \|\mathbf{x}\|_2^2)$.

Analysis of the (ε, δ) -JL Property (Cont.)

Define $z_i = \frac{w_i}{\|\mathbf{x}\|_2}$. Then, we have that

- $z_i \sim \mathcal{N}(0, 1)$;
- $z_i^2 = \frac{w_i^2}{\|\mathbf{x}\|_2^2}$;
- $\mathbb{E}[w_i^2] = \|\mathbf{x}\|_2^2$.

Besides, summing all k terms of w_i^2 together, we have:

$$\mathbb{E}\left[\frac{1}{k} \sum_{i=1}^k w_i^2\right] = \|\mathbf{x}\|_2^2.$$

Hence, we only need to care if the random variable $Y = \frac{1}{k} \sum_{i=1}^k w_i^2$ concentrates on its expectation or not. Rewrite Y as follows:

$$Y = \frac{1}{k} \sum_{i=1}^k w_i^2 = \|\mathbf{x}\|_2^2 \cdot \frac{\sum_{i=1}^k z_i^2}{k}.$$

Analysis of the (ε, δ) -JL Property (Cont.)

Definition 2 (χ^2 random variable)

A χ^2 -squared random variable χ_k^2 with k degrees of freedom can be represented as the sum $\sum_{i=1}^k z_i^2$ where z_i are i.i.d. from $\sim \mathcal{N}(0, 1)$.

Theorem 2 (Concentration bound of χ_k^2 random variables [2])

For a χ_k^2 random variable $\sum_{i=1}^k z_i^2$, the following bound holds:

$$\mathbb{P}\left[\left|\frac{\sum_{i=1}^k z_i^2}{k} - 1\right| \geq \varepsilon\right] \leq 2e^{-k \cdot \varepsilon^2 / 8}$$

Then, we can get that when $k = \frac{8 \log(2/\delta)}{\varepsilon^2}$,

$$\mathbb{P}\left[\left|\frac{\sum_{i=1}^k z_i^2}{k} - 1\right| \geq \varepsilon\right] \leq \delta \Leftrightarrow \mathbb{P}[|Y - \|\mathbf{x}\|_2^2| \geq \varepsilon \|\mathbf{x}\|_2^2] \leq \delta.$$

This finishes the proof.

Another Proof via Moment Generating Function

Lemma 4

For a χ_k^2 random variable $Z = \sum_{i=1}^k Z_i^2$, the following bound holds:

$$\mathbb{P}[Z > (1 + \varepsilon)^2 \cdot k] \leq e^{-\frac{3k \cdot \varepsilon^2}{4}}$$

Firstly, let $t \in (0, 0.5)$, we have that:

$$\mathbb{P}[Z > (1 + \varepsilon)^2 k] = \mathbb{P}[e^{tZ} > e^{(1+\varepsilon)^2 k \cdot t}] \leq \frac{\mathbb{E}[e^{tZ}]}{e^{(1+\varepsilon)^2 k \cdot t}} \quad (2)$$

Also note that:

$$\mathbb{E}[e^{tZ}] = \mathbb{E}[e^{t \sum_{i=1}^k z_i^2}] = \prod_{i=1}^k \mathbb{E}[e^{t \cdot z_i^2}].$$

Another Proof via Moment Generating Function (Cont.)

We consider $\mathbb{E}[e^{t \cdot z_i^2}]$ separately.

$$\begin{aligned}\mathbb{E}[e^{t \cdot z_i^2}] &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{ty^2} \cdot e^{-\frac{y^2}{2}} dy = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-\frac{(1-2t)y^2}{2}} dy \\ &= \frac{1}{\sqrt{2\pi}\sqrt{1-2t}} \int_{-\infty}^{+\infty} e^{-\frac{z^2}{2}} dz = \frac{1}{\sqrt{1-2t}}.\end{aligned}$$

Then, we have $\mathbb{E}[e^{tZ}] = (1-2t)^{-k/2}$. Putting it into Equation (2), we have:

$$\mathbb{P}[Z > (1+\varepsilon)^2 \cdot k] \leq e^{-(1+\varepsilon)^2 \cdot k \cdot t} \cdot (1-2t)^{-k/2}. \quad (3)$$

Choose the minimum of $f(t) = e^{-(1+\varepsilon)^2 k \cdot t} \cdot (1-2t)^{-k/2}$. To do this, we may first take $f_1(t) = \ln f(t)$ and then take the derivative on $f_1(t)$. We can derive that: $t = (1 - \frac{1}{(1+\varepsilon)^2})/2 \in (0, 0.5)$.

Another Proof via Moment Generating Function (Cont.)

Putting $t = (1 - \frac{1}{(1+\varepsilon)^2})/2$ into Equation (3), we have:

$$\mathbb{P}[Z > (1 + \varepsilon)^2 k] \leq e^{-(1+\varepsilon)^2 \cdot k \cdot t} \cdot (1 - 2t)^{-k/2} = e^{-k(\varepsilon + \varepsilon^2/2 - \ln(1+\varepsilon))}. \quad (4)$$

Using the fact that $\ln(1+x) \leq x - \frac{x^2}{4}$, we have that

$$\mathbb{P}[Z > (1 + \varepsilon)^2 k] \leq e^{-\frac{3k\varepsilon^2}{4}}.$$

This finishes the proof of Lemma 4

We also have the following theorem for the lower tail.

Lemma 5

For a χ_k^2 random variable $Z = \sum_{i=1}^k Z_i^2$, the following bound holds:

$$\mathbb{P}[Z < (1 - \varepsilon)^2 \cdot k] \leq e^{-\frac{k \cdot \varepsilon^2}{2}}$$

Alternative (ϵ, δ) -JL Construction

The previous solution uses a dense matrix for projection. Do we really need such a dense matrix to preserve the pairwise distance? The next theorem provides an alternative solution with almost $\frac{2}{3}$ entries of the projection matrix to be zero.

Theorem 3 (Achlioptas [3])

Let $\mathbf{\Pi}$ be a $k \times d$ matrix where each entry is i.i.d. drawn from the following distribution:

$$\Pi_{ij} = \begin{cases} \frac{\sqrt{3}}{\sqrt{k}} & \text{with probability } \frac{1}{6}, \\ 0 & \text{with probability } \frac{2}{3}, \\ -\frac{\sqrt{3}}{\sqrt{k}} & \text{with probability } \frac{1}{6}. \end{cases}$$

Then, $\mathbf{\Pi}$ has the (ϵ, δ) -JL property when $k = O(\frac{\log 1/\delta}{\epsilon^2})$.

Proof Roadmap

We prove the concentration of $\|\Pi \mathbf{x}\|_2^2$ around $\|\mathbf{x}\|_2^2$ in steps.

Lemma 6 (Unbiasedness)

Let $\pi_i^\top$ be row i of Π and $Y_i = \langle \pi_i, \mathbf{x} \rangle$. Then

$$\mathbb{E}[\|\Pi \mathbf{x}\|_2^2] = \sum_{i=1}^k \mathbb{E}[Y_i^2] = \|\mathbf{x}\|_2^2.$$

Definition 3 (Subgaussian random variable)

A (mean-zero) random variable X is *subgaussian* with parameter σ^2 if

$$\mathbb{E}\left[e^{tX}\right] \leq \exp\left(\frac{\sigma^2 t^2}{2}\right) \quad \forall t \in \mathbb{R}.$$

We are going to show that Y_i is a subgaussian random variable.

Proof Roadmap (Cont.)

Lemma 7 (Row inner product is subgaussian)

Let $\pi_i^\top$ be row i of $\Pi \in \mathbb{R}^{k \times d}$ and $Y_i = \langle \pi_i, \mathbf{x} \rangle$ for $\mathbf{x} \in \mathbb{R}^d$. Then Y_i is mean-zero and subgaussian with parameter $\sigma^2 = \|\mathbf{x}\|_2^2/k$, i.e.,

$$\mathbb{E}\left[e^{tY_i}\right] \leq \exp\left(\frac{\sigma^2 t^2}{2}\right) = \exp\left(\frac{t^2 \|\mathbf{x}\|_2^2}{2k}\right), \quad \forall t \in \mathbb{R}.$$

Define $Z_{ij} := \Pi_{ij} x_j$ so that $Y_i = \sum_{j=1}^d Z_{ij}$, and the Z_{ij} are independent over j . For any $t \in \mathbb{R}$, with $u := tx_j/\sqrt{k}$,

$$\mathbb{E}\left[e^{tZ_{ij}}\right] = \frac{2}{3} + \frac{1}{6}e^{\sqrt{3}u} + \frac{1}{6}e^{-\sqrt{3}u}.$$

We use the fact that

$$\frac{2}{3} + \frac{1}{6}e^{\sqrt{3}u} + \frac{1}{6}e^{-\sqrt{3}u} \leq e^{u^2/2} \quad (\forall u \in \mathbb{R}),$$

Proof Roadmap (Cont.)

Lemma 8 (Factorial-accurate even-moment bound)

Let $\pi_i^\top$ be row i of $\Pi \in \mathbb{R}^{k \times d}$ and $Y_i = \langle \pi_i, \mathbf{x} \rangle$ for $\mathbf{x} \in \mathbb{R}^d$. Then for every integer $p \geq 1$,

$$\mathbb{E}[Y_i^{2p}] \leq \frac{(2p)!}{2^p p!} \sigma^{2p}.$$

(In our setting) $\sigma^2 = \|\mathbf{x}\|_2^2/k$, hence

$$\mathbb{E}[Y_i^{2p}] \leq \frac{(2p)!}{2^p p!} \left(\frac{\|\mathbf{x}\|_2^2}{k} \right)^p.$$

Proof Roadmap (Cont.): MGF Bound for Y_i^2

Lemma 9 (MGF of the squared row inner product)

Let Y_i be mean-zero and sub-gaussian with parameter σ^2 . For $t < 1/(2\sigma^2)$:

$$\mathbb{E}[e^{tY_i^2}] \leq \frac{1}{\sqrt{1-2t\sigma^2}}$$

Proof.

Expand the MGF: Using the Taylor series for the exponential function:

$$\mathbb{E}[e^{tY_i^2}] = \mathbb{E}\left[\sum_{p=0}^{\infty} \frac{(tY_i^2)^p}{p!}\right] = \sum_{p=0}^{\infty} \frac{t^p}{p!} \mathbb{E}[Y_i^{2p}]$$



Proof of Lemma 9 (MGF bound for Y_i^2) (Cont.)

Proof.

Apply even-moment bound: From Lemma 8, $\mathbb{E}[Y_i^{2p}] \leq \frac{(2p)!}{2^p p!} \sigma^{2p}$:

$$\mathbb{E}[e^{tY_i^2}] \leq \sum_{p=0}^{\infty} \frac{t^p}{p!} \cdot \frac{(2p)!}{2^p p!} \sigma^{2p} = \sum_{p=0}^{\infty} \frac{(2p)!}{2^p (p!)^2} (t\sigma^2)^p$$

Comparison with Binomial Series: The term $\frac{(2p)!}{2^p (p!)^2}$ corresponds to the coefficients of the Taylor expansion for $(1-2x)^{-1/2}$. Specifically:

$$\frac{1}{\sqrt{1-2t\sigma^2}} = \sum_{p=0}^{\infty} \binom{-1/2}{p} (-2t\sigma^2)^p = \sum_{p=0}^{\infty} \frac{(2p)!}{2^p (p!)^2} (t\sigma^2)^p$$

Hence, for $2t\sigma^2 < 1$, we have that: $\mathbb{E}[e^{tY_i^2}] \leq \frac{1}{\sqrt{1-2t\sigma^2}}$



Proof Completion: Achlioptas Construction

Let $Z = \sum_{i=1}^k Y_i^2$. Since each row has variance $\sigma^2 = \|x\|^2/k$:

$$\mathbb{E}[e^{tZ}] = \prod_{i=1}^k \mathbb{E}[e^{tY_i^2}] \leq (1 - 2t\sigma^2)^{-k/2}$$

Setting $\mu = \|x\|^2 = k\sigma^2$, we bound the upper tail:

$$\mathbb{P}[Z > (1 + \varepsilon)^2 \mu] \leq e^{-t(1+\varepsilon)^2 k\sigma^2} (1 - 2t\sigma^2)^{-k/2}$$

The minimum occurs at $t = \frac{1}{2\sigma^2} (1 - \frac{1}{(1+\varepsilon)^2})$. Putting back:

$$\mathbb{P}[Z > (1 + \varepsilon)^2 \mu] \leq e^{-k(\varepsilon + \varepsilon^2/2 - \ln(1+\varepsilon))}$$

It is the same as our previous analysis for Gaussian matrix setting. So the remaining proof is omitted. Setting $k = O\left(\frac{\log(1/\delta)}{\varepsilon^2}\right)$ provides the approximation guarantee.

Theorem 4 (Kane and Nelson[4])

There exist distributions on $\Pi \in \mathbb{R}^{k \times d}$ such that, when $k = O(\log n / \varepsilon^2)$ and each row of dimension d includes $O(\frac{\log n}{\varepsilon})$ non-zero entries, the (ε, δ) -JL property is satisfied.

Application of JL-Lemma

- Approximate near neighbor search (ANNS) under L_2 -norm. First, reduce the dimension d to k and then apply the ANNS solution, e.g., tree structure, LSH, Product Quantization, and graph-based indices, on the newly mapped space.
- Approximate linear regression. Given a set of data points $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ in $\mathbb{R}^d$ and corresponding target values $\mathbf{y} = \{y_1, y_2, \dots, y_n\}$. The goal is to find $\mathbf{a} \in \mathbb{R}^d$ such that:

$$\min_{\mathbf{a} \in \mathbb{R}^d} \sum_{i=1}^n (\mathbf{x}_i \cdot \mathbf{a} - y_i)^2 = \min_{\mathbf{a} \in \mathbb{R}^d} \|\mathbf{X}\mathbf{a} - \mathbf{y}\|_2$$

By JL Lemma, we can solve the approximate version:

$$\min_{\mathbf{a} \in \mathbb{R}^d} \|\Pi \mathbf{X} \mathbf{a} - \Pi \mathbf{y}\|_2.$$

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